Circuit Documentation

Summary

This circuit is designed to control multiple AC bulbs using a NodeMCU microcontroller and a 4 Channel Relay Module. The circuit includes several rocker switches for manual control, a push button for resetting the NodeMCU, and a power supply module to convert AC to DC for powering the microcontroller and relay module. The AC bulbs are controlled through the relay module, which is interfaced with the NodeMCU. The circuit is powered by an AC supply connected through an MCB for safety.

Component List

1. **Rocker Switch (SPST)**
   - Description: Single Pole Single Throw switch used for manual control.
   - Pins: 1, 2
2. **Hilink 220V to 5V**
   - Description: Power supply module to convert 220V AC to 5V DC.
   - Pins: AC/L, AC/N, -Vo, +Vo
3. **Push Button**
   - Description: Used to reset the NodeMCU.
   - Pins: pin1, pin2
4. **AC Bulb**
   - Description: Standard AC bulb used as a load.
   - Pins: Hot, Neutral
5. **4 Channel Relay Module**
   - Description: Relay module used to control AC loads.
   - Pins: GND, IN1, IN2, IN3, IN4, VCC, NC1, COM1, NO1, NC2, COM2, NO2, NC3, COM3, NO3, NC4, COM4, NO4
6. **NodeMCU**
   - Description: Microcontroller used to control the relay module.
   - Pins: A0, RSV, SD3, SD2, SD1, CMD, SD0, CLK, GND, 3V3, EN, RST, Vin, D0, D1, D2, D3, D4, D5, D6, D7, D8, RX, TX
7. **AC MCB (Blue)**
   - Description: Miniature Circuit Breaker for safety.
   - Pins: LIVE IN, NEUTRAL IN, LIVE OUT, NEUTRAL OUT
8. **AC Supply**
   - Description: Provides AC power to the circuit.
   - Pins: +ve, -ve

Wiring Details

Rocker Switch (SPST)

- **Switch 1**: Pin 1 connected to NodeMCU D5, Pin 2 connected to NodeMCU GND.
- **Switch 2**: Pin 1 connected to NodeMCU D6, Pin 2 connected to NodeMCU GND.
- **Switch 3**: Pin 1 connected to NodeMCU D7, Pin 2 connected to NodeMCU GND.
- **Switch 4**: Pin 1 connected to NodeMCU D8, Pin 2 connected to NodeMCU GND.

Hilink 220V to 5V

- AC/L connected to AC MCB LIVE OUT.
- AC/N connected to AC MCB NEUTRAL OUT.
- -Vo connected to NodeMCU GND and Relay Module GND.
- +Vo connected to NodeMCU Vin and Relay Module VCC.

Push Button

- Pin 1 connected to NodeMCU RST.
- Pin 2 connected to NodeMCU GND.

AC Bulb

- **Bulb 1**: Hot connected to Relay Module NC1, Neutral connected to AC MCB NEUTRAL OUT.
- **Bulb 2**: Hot connected to Relay Module NC2, Neutral connected to AC MCB NEUTRAL OUT.
- **Bulb 3**: Hot connected to Relay Module NC3, Neutral connected to AC MCB NEUTRAL OUT.
- **Bulb 4**: Hot connected to Relay Module NC4, Neutral connected to AC MCB NEUTRAL OUT.

4 Channel Relay Module

- IN1 connected to NodeMCU D1.
- IN2 connected to NodeMCU D2.
- IN3 connected to NodeMCU D3.
- IN4 connected to NodeMCU D4.
- COM1, COM2, COM3, COM4 connected to AC MCB LIVE OUT.

NodeMCU

- GND connected to Hilink -Vo and Relay Module GND.
- Vin connected to Hilink +Vo.
- D1, D2, D3, D4 connected to Relay Module IN1, IN2, IN3, IN4 respectively.
- D5, D6, D7, D8 connected to Rocker Switches 1, 2, 3, 4 respectively.

AC MCB (Blue)

- LIVE IN connected to AC Supply +ve.
- NEUTRAL IN connected to AC Supply -ve.
- LIVE OUT connected to Relay Module COM pins and Hilink AC/L.
- NEUTRAL OUT connected to AC Bulbs Neutral and Hilink AC/N.

This documentation provides a comprehensive overview of the circuit, detailing each component, its connections.